Systems Analysis

IS2101 -

What is Systems Analysis?

- Systems Analysis is the study of a system and its components.
- It is a prerequisite to Systems Design.
- Systems Analysis is driven by the business concerns of ‘System owners’ and ‘System Users’.
- It addresses the knowledge, process and communication building blocks from systems owners’ and users’ perspectives.
- ‘**Systems Analyst**’ serve as facilitators for systems analysis.

Systems Analyst Roles and Skills

- Project team needs a variety of skills
- Project members identify ways to improve an organization, build an information system to support them, and train and motivate others to use the system.
- Understanding what to change and how to change it—and convincing others of the need for change—requires a wide range of skills.
- Analyst Skills in six major categories:
  - Technical
  - Business
  - Analytical
  - Communicational
  - Management
  - Ethical

Systems Analyst Skills cont.

- **Analytical skills**
  - Analysts are continuous problem solvers
  - Ability to see things as systems, identify, analyze, and solve problems in an optimal way for a specific organization.
- **Technical skills** –
  - to understand the organization's existing technical environment, technology that will make up the new system, the way both can fit into an integrated technical solution
  - ability to understand how computers, data networks, databases, operating systems, etc. work together, as well as their potentials and limitations.
- **Business skills** - to understand how IT can be applied to business situations and to ensure that the IT delivers real business value.

Systems Analyst Skills cont..

- **Communication skills** –
  - communicate effectively **one-on-one** with users and business managers (-little experience with technology) and with programmers (-more technical expertise than the analyst).
  - able to give presentations to large and small groups and write reports
  - need to have strong interpersonal abilities
  - include effective interpersonal communication (written, verbal, visual, electronic, face-to-face conversations, presentations in front of groups), listening, group facilitation skills.
- **Management skills**
  - include organization's recourse management, project management (people and money), risk management, and change management.
  - need to manage people with whom they work and they need to manage the pressure and risks associated with unclear situations

Systems Analyst Skills cont..

- **Ethical**
  - Analysts must deal fairly, honestly, and ethically with other project team members, managers, and system users.
  - Analysts often deal with confidential information or information that, if shared with others, could cause harm (e.g., conflict among employees); it is important to maintain confidence and trust with all people.
- In addition to these six general skill sets, analysts require many specific skills associated with roles performed on a project.
- Most large organizations build project teams containing several individuals with clearly defined responsibilities. – programmers, technical writers etc.
- Different organizations divide the roles differently.

Systems Analyst Skills - Analytical Skills

- **Analytical skills - 4 sub skills**
  1. Systems Thinking
  2. Organizational Knowledge
  3. Problem Identification
  4. Problem Analyzing and Solving

Analytical Skills - Systems Thinking

- Identifying something as a system, visualizing the system and translating it into abstract terms, and thinking about the characteristics of specific situation.
- Information systems can be seen as subsystems in larger organizational systems, taking input from, and returning output to, their organizational environments.

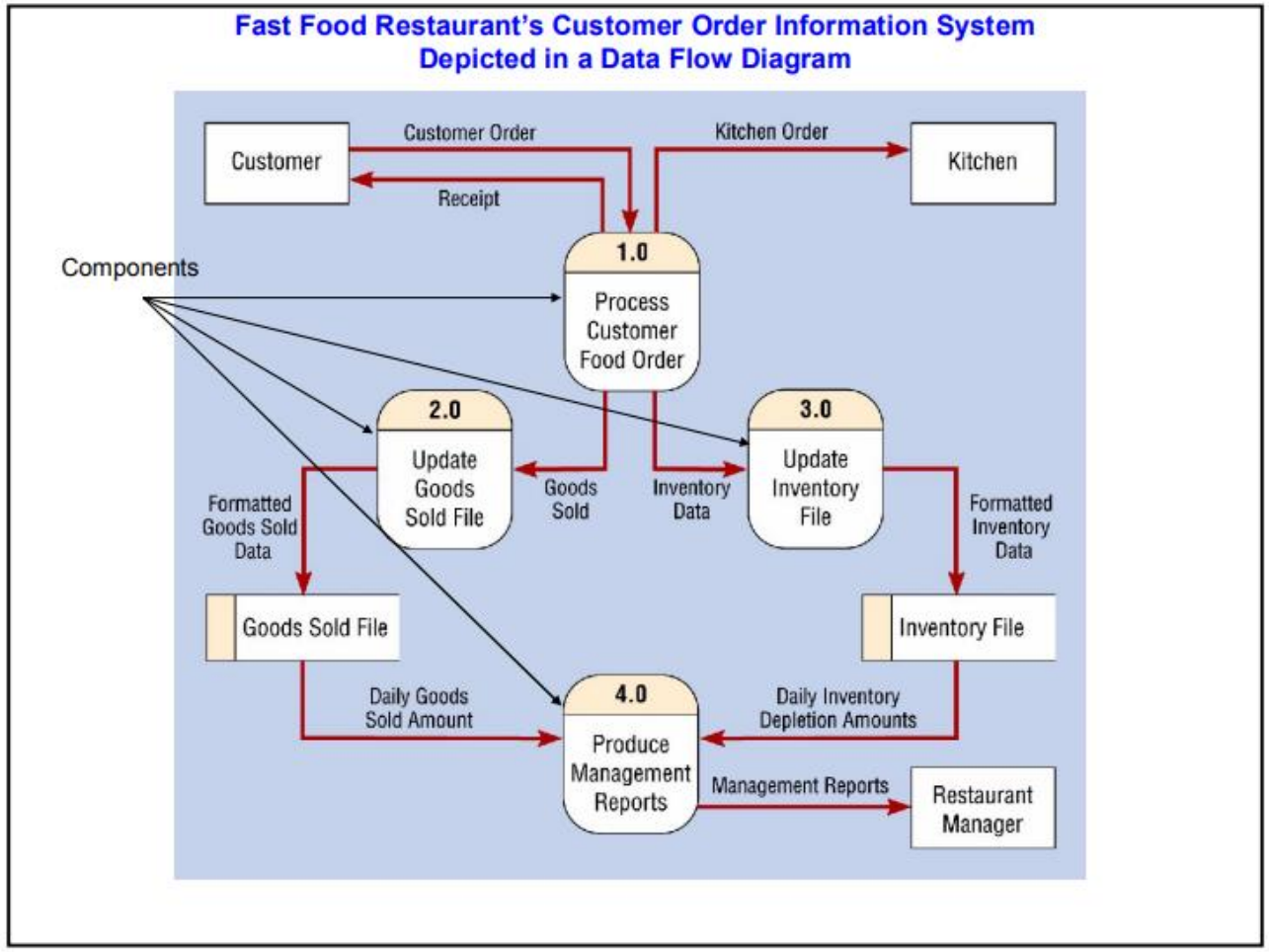
Analytical Skills- Organization Knowledge

- System analyst must understand how organization works.
- Organizations are systems because they are made up of interrelated components working together for a purpose.
- They take input from and return output to their environments.
- Organizations can be redesigned through a systems analysis and design process to replace system components while preserving interconnections between components.
- Examples of Organizational Knowledge: (virtually any piece of information will help)
  - how work officially gets done in a particular organization,
  - understanding of organization rules and regulations, internal policies,
  - understanding of org. competitive and regulatory environments,
  - understanding of org. strategies and tactics,

Definition of a System and Its Parts

- **A System** - is an inter-related set of components, with an identifiable boundary, working together for some purpose.
- Characteristics:
  - **Components** - an aggregation of parts
  - **Relations** (links) - dependence of one subsystem on one or more subsystems
  - **A Boundary** - a line that sets off the system from its environment
  - **A Purpose** - the overall goal or function of a system
  - **An Environment** - everything external to a system that interacts with the system
  - **Interfaces** - points of “system-environment” and “subsystem- subsystem” contacts
  - **Input** - whatever a system takes from its environment
  - **Output** - whatever a system returns to its environment
  - **Constraint(s)** - limit(s) to what a system can accomplish

Fast Food Restaurant's Customer Order Information System Data Flow diagram



System concepts

- **Open system**: A system that interacts freely with its environment, taking input and returning output.
- **Closed system**: a system that is cut off from its environment and does not interact with it.
- **Modularity :** dividing a system into parts/chunks/modules of relatively uniform size.
- **Decomposition** : process of breaking down a system into its component parts.
- **Coupling** : Extent to which subsystems are dependent on each other.
- **Cohesion** : Extent to which a system or a subsystem performs a single function.

Analytical Skills: Problem Analyzing and Solving

- In **problem identification**, system analyst MUST BE ABLE to compare the current situation in an organization to the desired situation: advantages and disadvantages of both, perspectives, etc.
- **Problem identification** involves measurement not decision making (identify and measure many details of a problem and create a solid foundation for a well-thought and weighted decision(s) and/or alternatives)
- **Problem solving** is the process of finding one or more ways to create the desired situation and select one approach for implementation.

Systems Analyst Skills - Technical Skills

- Computers (PCs, mini, mainframes, etc.)
- Computer networks (LAN, WAN, VPNs, administration, security, etc.)
- Operating systems (Unix, Mac/OS, Windows)
- Data Exchange Protocols (ftp, http, etc.)
- Programming languages (C++, Java, XML, etc.)
- Software applications (Office, project managements, etc.)
- Information systems (databases, MISs, decision support systems, etc.)
- System development tools and environments (report generators, office automation tools, etc.)

Systems Analyst Skills - Communication Skills

- Communication skills needed by systems analysts include
  1. Clear and effective interpersonal communication, written, verbal, or visual, from writing reports to face–to–face conversations, to presentations in front of groups
  2. listening (accepting opinions and ideas from other project team members),
  3. group facilitation skills:
  - setting an agenda,
  - leading discussions,
  - involving all parties in the discussion,
  - summarizing ideas,
  - keeping discussions on the agenda,

Roles: Business Analyst/ Business Systems Analyst(BSA)

- A business analyst focuses on the business issues surrounding the system.
- These issues include
  - identifying the business value that the system will create
  - developing ideas and suggestions for how the business processes can be improved
  - designing the new processes and policies in conjunction with the systems analyst.
- This individual likely has business experience and some type of professional training.
- He or she represents the interests of the project sponsor and the ultimate users of the system.
- A business analyst assists in the planning and design phases but is most active in the analysis phase.

Roles: Systems Analyst

- A systems analyst focuses on the IS issues surrounding the system.
- Develops ideas and suggestions for,
  - how information technology can improve business processes
  - designs the new business processes with help from the business analyst
  - designs the new information system, and ensures that all IS standards are maintained
- A systems analyst likely has significant training and experience in analysis and design, programming, and even areas of the business.
- represents the interests of the IS department and works intensively through the project but perhaps less so during the implementation phase.

Roles: Infrastructure Analyst

- An infrastructure analyst focuses on the technical issues surrounding how the system will interact with the organization's technical infrastructure (e.g., hardware, soft ware, networks, and databases).
- An infrastructure analyst's tasks include ensuring that the new information system conforms to organizational standards and identifying infrastructure changes needed to support the system.
- This individual probably has significant training and experience in networking, database administration, and various hardware and software products.
- He or she represents the interests of the organization and IS group that will ultimately have to operate and support the new system once it has been installed.
- An infrastructure analyst works throughout the project but perhaps less so during planning and analysis phases.

Roles: Change Management Analyst

- A change management analyst focuses on the people and management issues surrounding the system installation.
- The roles include ensuring that the adequate documentation and support are available to users, providing user training on the new system, and developing strategies to overcome resistance to change.
- This individual should have significant training and experience in organizational behavior in general and change management in particular.
- A change management analyst works most actively during the implementation phase but begins laying the groundwork for change during the analysis and design phases.

Systems Analyst as a profession:

- Generally accepted characteristics of a profession :
  - college curriculum guidelines,
  - standards of practice,
  - professional societies,
  - certification programs,
  - job descriptions,
  - professional training,
  - codes of ethics.
- A code of ethics is a list of statements of intended conduct, emphasizing personal responsibility, honesty….
- https://www.prospects.ac.uk/job-profiles/systems-analyst

Systems Analysis Approaches

- Systems Analysis is about Problem solving
- Different approaches:
  1. Model driven Analysis Approaches
  2. Accelerated Systems Analysis Approaches

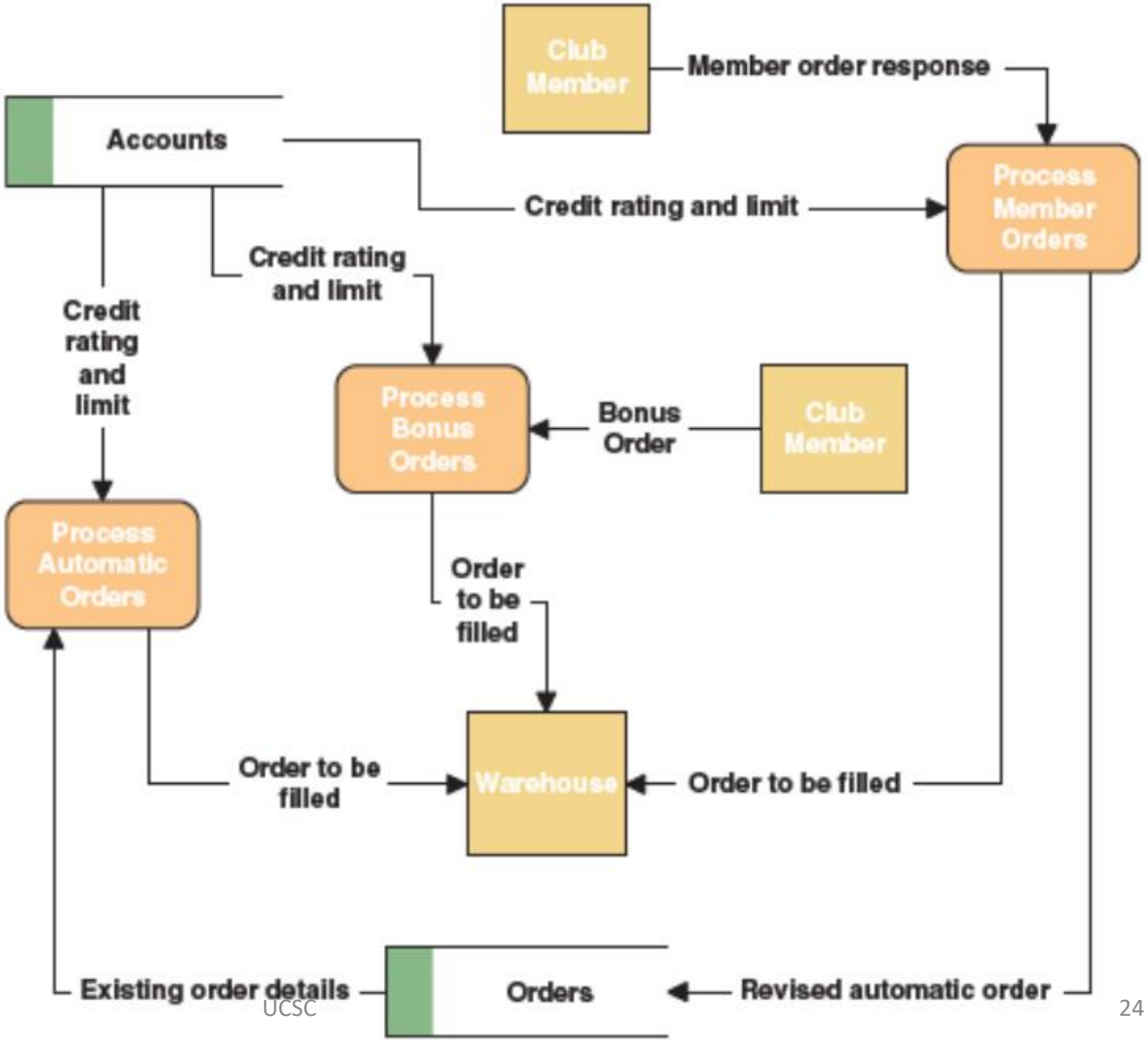
1. Model driven Analysis Approaches

- E.g. Structured Analysis, Information engineering, Object oriented analysis
- Use pictures to communicate business requirements and solutions.
- Enhanced by the use of automated tools.- CASE or modelling tools
- CASE tools offer the advantage of consistency and completeness analysis as well as rule-based error checking.
- Featured in model-driven methodologies.
- Examples:
  - Structured analysis
  - Information Engineering (IE)
  - Object oriented approach

Structured analysis

- Focuses on flow of data through business and software processes
- Process centered - emphasis on the processes
- Key tool to model processes are ‘Data flow diagrams’
- Data flow diagrams- existing and/or proposed processes, inputs, outputs and data, show the flow of data between and through processes and show the places where data is stored
  - Serve as blueprints for business processes to be implemented and software to be purchased or constructed

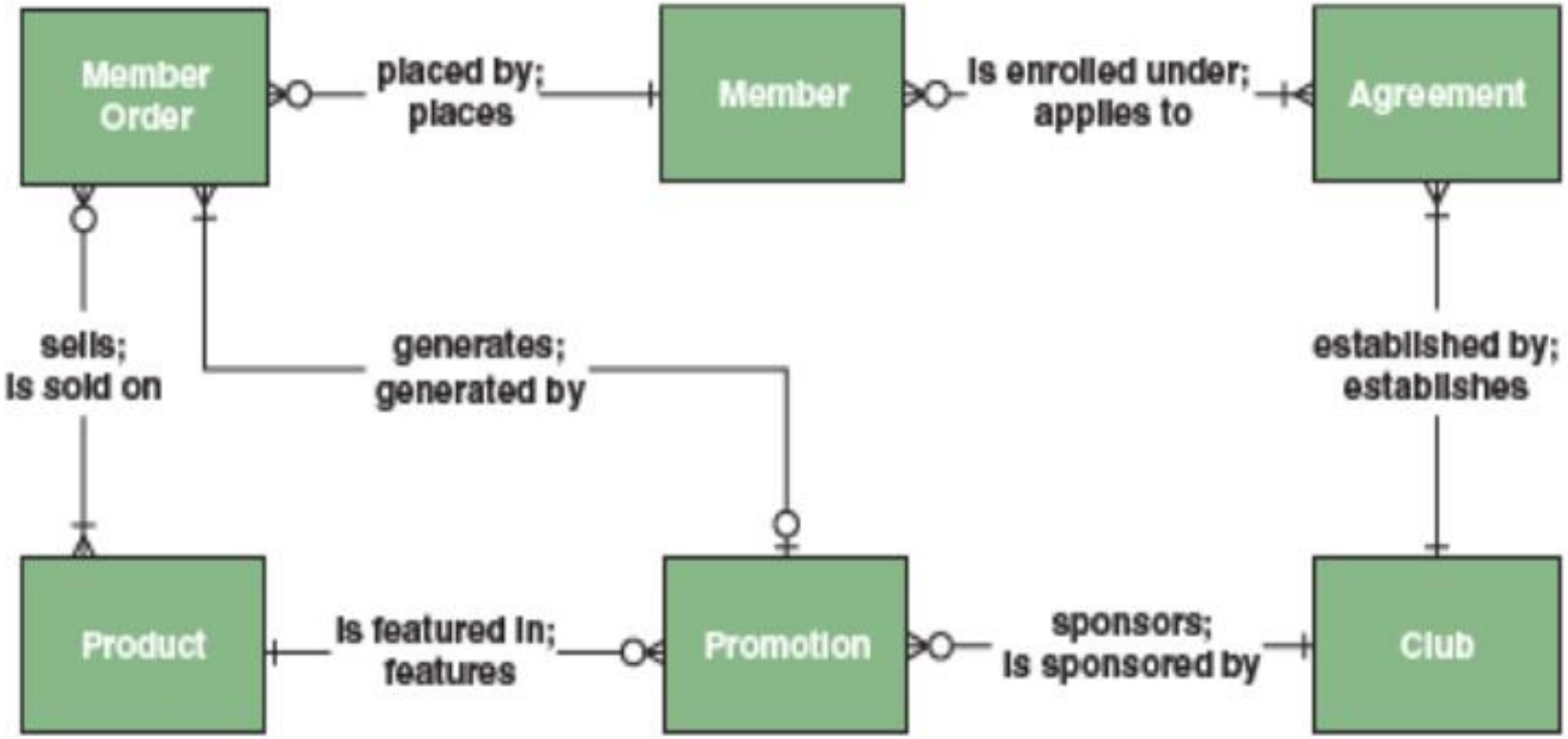
Process Model (Data flow model)



Information Engineering (IE)

- Focus on structure of stored data
- Data centered, emphasizing the analysis of knowledge(data) requirements
- Key tool- Entity Relationship diagram(ERD)
- IE and Structured Analysis ae complementary approaches
  - use data flow dL1grams to model a system's processes and entity relationship diagrams to model a system’s data

Data model/ERD



Object oriented approach

- Emerged to eliminate the separation of processes and data.
- Consider Information systems as a collection of **objects** that encapsulate data and processes.
- Objects can contain data attributes
- But to create, read, update, or delete an object's data is through its embedded processes
- OO programming languages – C++ , java, .NET

2. Accelerated Systems Analysis Approaches

- E.g. **Discovery prototyping, Rapid Architected Development**
- construction of prototypes to more rapidly identify business and user requirements for a new system
- Prototype - working but incomplete samples of a desired system
  - ‘I will know what I want when I see it” - Characteristic of many users and managers
  - Incomplete- prototype will not include error checking, input data validation, security, and processing completeness of a finished application.
  - Can develop quickly – identify most crucial of business level requirements
- Evolutionary prototyping
- Common in rapid application development(RAD) methodologies

Discovery prototyping

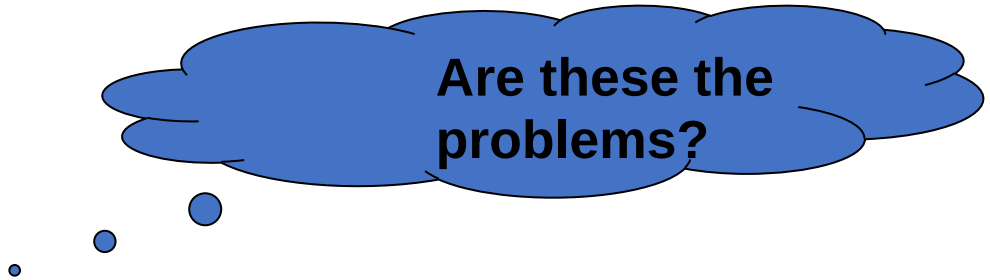
- Use rapid development technology to help users discover their business requirements.
- E.g. using MS Access to create simple database, forms and reports
- Final systems develop in a more sophisticated application development tool and language
- Simpler tool allows the analyst to more quickly prototype the user's requirements
- Problems :
  - Preoccupy with the final 'look and feel' of the prototype systems
  - Users be misled to believe that tools like Access can be used to build the final system

Rapid Architected Analysis

- Accelerated analysis approach that also builds system models
- Made possible by **Reverse- Engineering technology** included in many automated tools such as CASE and programming languages
- Reverse Engineering tools generate system models from existing software applications – models for new and improved systems
- Resulting system models can then be edited and Improved by systems analysts
- Blend of Model driven and Accelerate analysis approaches
- 2 different techniques for applying rapid architected analysis:
  - Legacy information systems - reverse engineer into various system models.
  - Reverse engineering the prototypes, forward engineering the models

Finding Problems to Solve

- Requirements solve problems
  - E.g. A mother takes her young daughter to the doctor because the child is ill. The first thing the doctor tries to do is, identify the problem.
    - The child has
      - earache,
      - fever,
      - runny nose.



Finding Problems to Solve

- E.g. Cont…
  - Mother has given the child pain killers to ease the pain,
  - But still the child is not well.
  - The mother is treating
    - the symptoms and
    - **NOT** the real problem.
  - However, the doctor,
    - analyze the symptoms further
    - examine the child
    - Make the conclusion

Conclusion (Root cause of the child's symptoms) : AN EAR INFECTION

Finding Problems to Solve

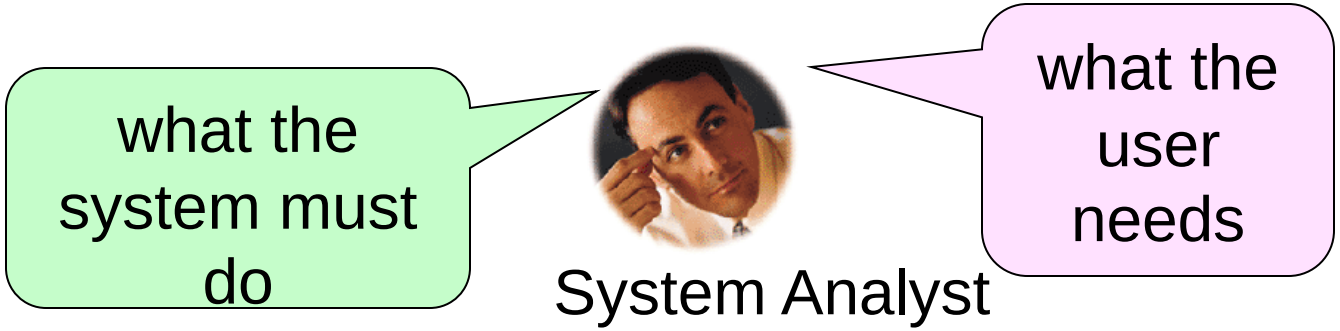
- E.g. Cont…
  - Problem is identified and analyzed,
  - Recommend a cure (solution)
    - An antibiotic can be prescribed
    - Determine constrains on the medicine that can prescribe.
      - How old is the child?
      - How much does she weigh?
      - Is the child allergic to any medications?
      - Can she swallow pills?
    - Once the constraints are known, a prescription can be generated.

- Systems analysts use the same problem-solving process as a doctor uses, but instead of diagnosing medical problems they diagnose system problems.

Requirements Analysis

Identifying Requirements

- Useful systems can only be built if you know exactly



- Therefore most important factors in building correct systems is to first clearly define what the system must do

Requirements Analysis

- **System Requirements**

Defines the services the system is to provide and prescribe constraints for its operation

Functional Requirements

Eg. Lending in a Library system

Non functional Requirements

Eg. Security,Performance

Importance of Communication

- Analyst must ensure that no ambiguities arise in discussions between various people involved in analysis
- Different jargon used by different people may cause problems

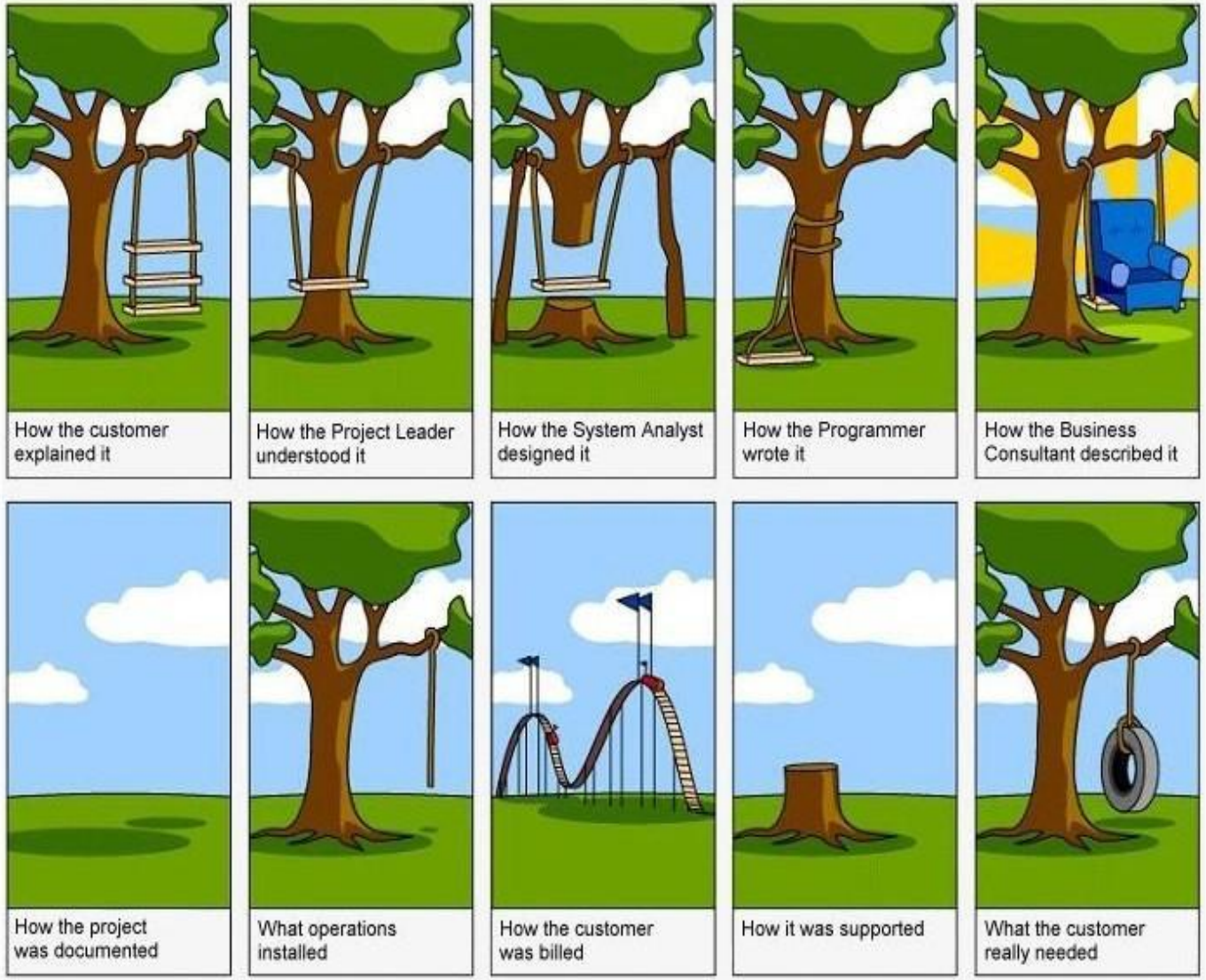
Example: Ambiguous Requirement Statement

Identify a transportation to transfer a single individual from home to place of work

Management Interpretation

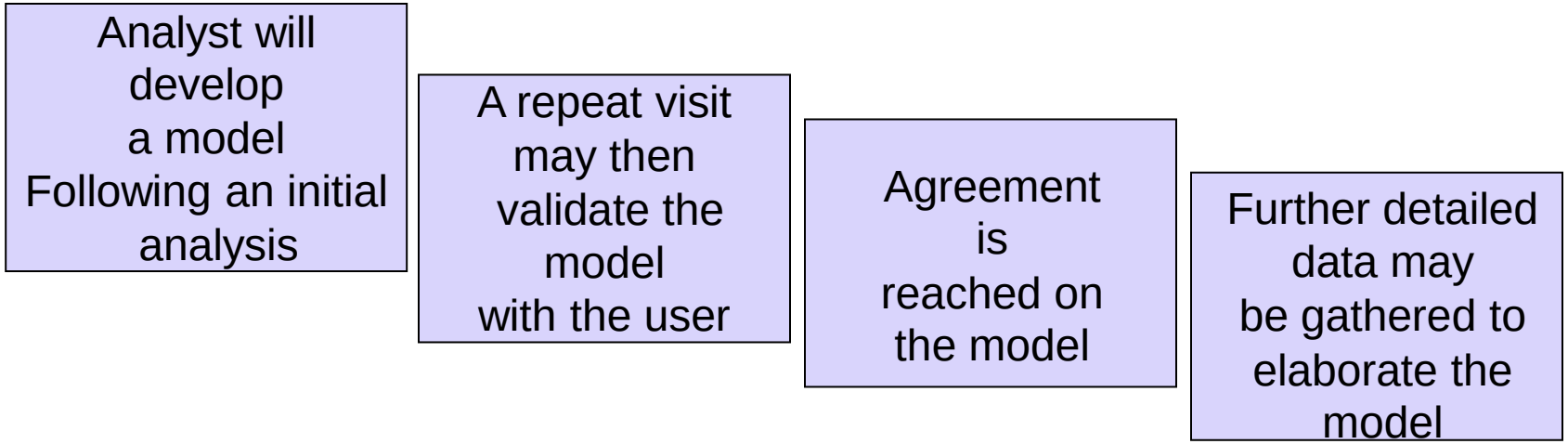
IT Interpretation

User Interpretation



The ugly fact about requirements engineering

Identifying Requirements



This iterative approach serves a number of purposes:

- there is always a record of information gathered to date
- ensures correctness of the information as you continually verify the results with the user.
- Analyst does not get too far ahead using wrong assumptions.

The requirements Analysis

- Definition phase or Logical design phase
- What do the users need and want from a new system?
- New systems will always be evaluated on whether or not they fulfill business objectives and requirements.
- Requirements analysis phase typically includes following tasks:
  1. Identify and express system requirements
  2. Prioritize system requirements
  3. Update or refine the project plan
  4. Communicate the requirements statement
-

Identify and Express System Requirements

- The initial task of the retirements analysis phase.
- It is often the source of many errors, omissions, and conflicts.
- functional and nonfunctional requirements need to meet the objectives are identified.
- Ftmctlon.al requirements are frequently Identified ln terms of inputs, outputs, processes, and stored data that are needed to satisfy the sy
- nonfunctional requirements Include performance (throughput and response time); ease of learning and use; budgets, costs, and cost savings; timetables and deadlines; documentation and training needs; quality management; and security and internal auditing controls. stem Improvement objectives
- SYSTEMS ANALYSIS fascilitate the task.

Prioritize system requirements

- If a projects delays or over budget, it is useful to recognize which requirements are more important than others.
- System owners and users should prioritize system requirements.
- Time boxing - timeboxing attempts to divide requirements into “chunks“’ that can be implemented within a period
- Time boxing forces priorities to be dearly defined.
- System analyst facilitate the prioritization task. System owners and users establish the actual priorities

Update or Refine the project plan

- Project scope is a moving target.
- Update project plan accordingly.
- If the new system larger than the originally expected team must adjust the schedule, budget, scope accordingly.
- Project manager, system owner and entire project team involve in this.

Communicate the requirements document

- Communication is an ongoing task of the requirements analysis phase.
- Communicate requirements and priorities to the business community throughout the phase.
- Communication is the process through which differences of opinion must be mediated.
- Project manager and sponsor should jointly facilitate this task.
- Interpersonal skills and negotiation skills are essential to this task.

Requirements Discovery Methods

Requirements Discovery

- Process and techniques used by systems analysts to identify, analyze and understand system requirements.
- Systems analyst work with system owners and users.

Requirement discovery for a system depends on the analysts' ability to first discover and then analyze problems and opportunities that exist in the current system

System Analyst

- Failure to correctly Identify system requirements may result in one or more of the following
- The system may cost more than projected.
- The system may be delivered later than promised.
- The system may not meet the users expectations, and that dissatlsfaction may cause them not to use it
- Once ln production, the costs of maitaing and enhancing the system may be excessively high.
- The system may be unreliable and prone to errors and downtime.
- ..

TABLE 6-2 Relative Coasts of Fixing an Error

Phase in which Error Discovered	Cost Ratio
Requirements	1
Design	3–6
Coding	10
Development Testing	15–40
Acceptance Testing	30–70
Operation	40–1,000

- In defining system requirements, it is critical that they meet the following criteria:
  - **Consistent-** The requirements are not conflicting or ambiguous.
  - **Complete-** The requirements describe all possible system inputs and responses.
  - **Feasible-** The requirements can be satisfied based on the available resources and constraints
  - **Required-** The requirements are truly needed and fulfill the purpose of the system.
  - **Accurate-** The requirements are stated correctly.
  - **Traceable-** The requirements directly map to the functions and features of the system
  - **Verifiable-** The requirements are defined so that they can be demonstrated during testing.

Requirements Discovery Methods

- Both model driven and accelerated systems analysis approaches attempt to express user requirements for a new system, either as models or as prototypes.
- Approaches to requirements discovery:
  - Fact-finding techniques
  - Joint requirements planning

Requirements Discovery Methods..

Fact-Finding

It is the formal process of using techniques to collect information about systems requirements etc. It is also called ***Information gathering***

Fact-Finding Methods

Sampling of Existing documents

Research and site visits

Observations of the work environment

Questionnaires

Interviews

Joint requirements planning

Prototyping

Fact finding ethics

- During fact-finding, systems analysts often come across or analyze information that is sensitive in nature.
- Analysts must take great care to protect the security and privacy of any facts or data with which they have been entrusted.
- Ethics policies document expected required behavior.
- Violations of these policies could lead to disciplinary action or even termination.
- Ethics play a crucial role in fact finding.

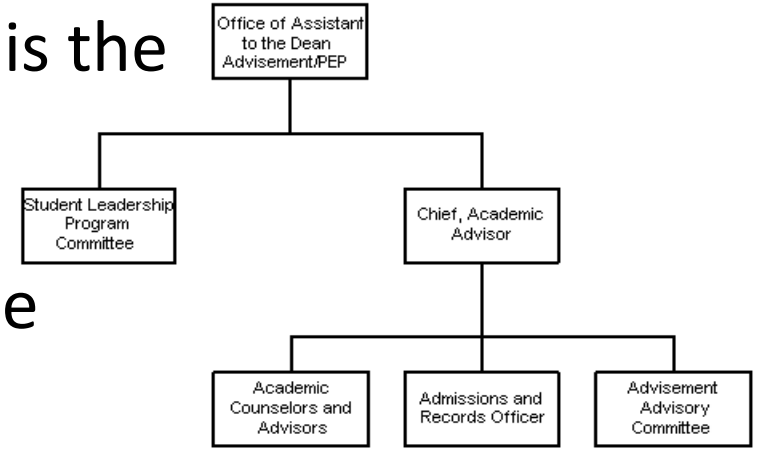
Requirements Discovery Methods..

Sampling of existing Documentation

- When you are studying an existing system, you can get a good idea by studying existing

Documentation Forms Files

- First document that analyst should seek out is the organizational chart



- Be sure that they are relevant and up-to-date

Requirements Discovery Methods..

Research and Site Visits

- Thoroughly research the application and problem.

Internet

Intranets

Computer trade Journals

Reference books

Good sources of information

World Wide Web

- Identify the material that are relevant and reliable

Requirements Discovery Methods..

Observations of the work environment

SA

- Systems Analyst participates in or watches a person perform activities to learn about the system

- Often used when validity of data collected through other methods is in question ***or*** when the complexity of certain aspects of the system prevents a clear explanation by the end users.

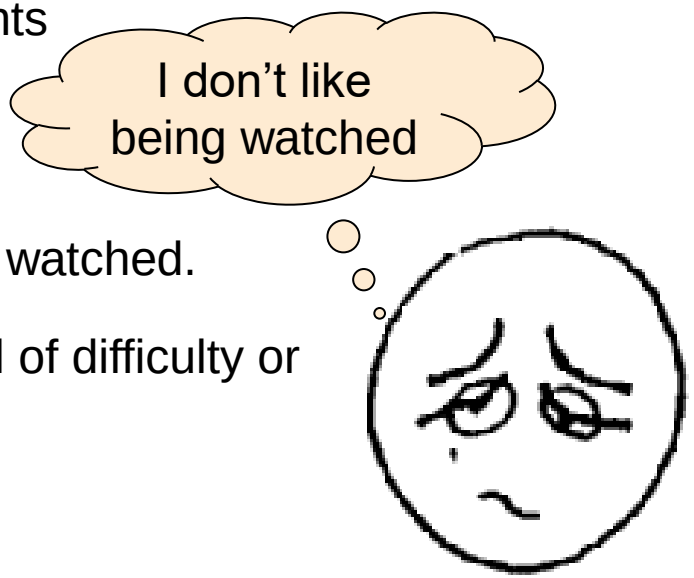
Requirements Discovery Methods..

Observations of the work environment…

Advantages

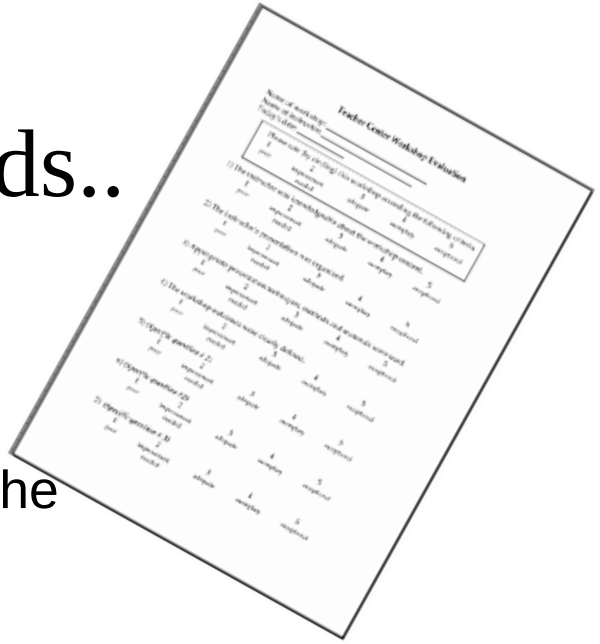
- Data gathered by observation can be highly reliable
- Relatively inexpensive
- Allows system analyst to do work measurements
- Etc.

Disadvantages



- People usually feel uncomfortable when being watched.
- Work being observed may not involve the level of difficulty or volume normally experienced during that time
- Etc.

Requirements Discovery Methods..

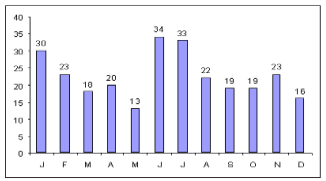


Questionnaires

Questionnaires are special purpose documents that allow the analysts to collect information and opinions from a large audience.

Advantages :

- Most questionnaires can be answered quickly
- Allow individuals to maintain anonymity
- Relatively inexpensive way of gathering data.
- Responses can be tabulated and analyzed quickly *etc.*



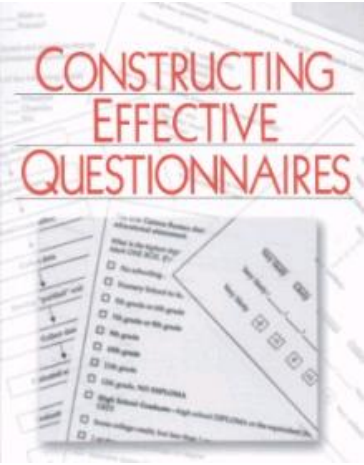
Requirements Discovery Methods..

Questionnaires…

Disadvantages:

- The number of respondents is often low
- Mostly suited for close questions
- No guarantee that an individual will answer or expand on all the questions
- Good Questionnaires are difficult to prepare
- No immediate opportunity to clarify a vague or incomplete answer to any question.

etc.



Requirements Discovery Methods..

Questionnaires…

Types of Questionnaires

- *Free-format* : A question is asked, and the respondent records the answer in the space provided after the question.
- *Fixed-format* : contains questions that require specific responses from individuals

There are 3 types of fixed-format questions

1. *multiple-choice questions*
   -Given several answers to select one. Eg. Yes, No type
2. *rating questions*
   -Given a statement and asked to use supplied responses to state an opinion.
3. *ranking questions*
   -Given a several possible answers to be ranked in order of preference or experience

- **Rating questions**

  a) Strongly agree

  b) Agree

  c) No opinion

  d) Disagree

  e) Strongly Disagree

- **Ranking Question**

  Rank the following Transactions according to the amount of time you spend processing them.

  - New Customer Orders
  - Order Cancellation
  - Order Modification
  - Payments

Requirements Discovery Methods..

Interviews

- Most commonly used technique in analysis
- Systems analyst collects information from individuals face to face.
- Systems Analyst must possess **good human relations skills** for dealing effectively with different type of people

Can be used to achieve any of the following goals:

find facts, verify facts, clarify facts, generate enthusiasm, get the end-user involved, identify requirements, and solicit ideas and opinions.

Requirements Discovery Methods..

Interviews…

Advantages

- Gives the analyst an opportunity to motivate the interviewee to respond freely and openly to questions.
- Allow the analyst to look for more feedback from the interviewee.
- Permit the analyst to ask questions from each individual etc.
- New ideas may arise

Motivation

Requirements Discovery Methods..

Interviews…

Disadvantages

- Very time consuming. Therefore, costly approach

- Success of interviews is highly dependent on the systems analyst's human relations skill.

- Interviews may be impractical due to the location of interviewees (If Video Conferencing/Teleconferencing not available)

Requirements Discovery Methods..

Interviews…

Types of Interviews

- Unstructured interviews .. conducted with only a general goal or subject in mind
- Structured interviews - has a specific set of questions to ask from the interviewee.

Types of Interview Questions

- Closed-ended questions : restrict answers to specific choices or short answers
- Open-ended questions : allow the interviewee to respond in any way that seems appropriate.

Requirements Discovery Methods..

Interviews…

How to conduct an *Interview*?

Select Interviewees

- Interview the end users of the information system you are studying.

- A formal organizational chart will help you identify these individuals and their responsibilities.

- Always make an appointment with the interviewee.

- Higher the management level of the interviewees, less time should be spent.

Requirements Discovery Methods..

Interviews…

How to conduct an *Interview*?...

Prepare for the Interview

Prepare an *interview guide* - checklist of specific questions interviewer will ask the interviewee

Avoid the type of questions such as:

- *Loaded* questions (Do you need to include both of these columns for this report?)
- *Leading* questions (You are not going to use this operator code, are you?)
- *Biased* questions (How many codes do we need for food classification in the inventory file? I think 20 should cover it ?)

Requirements Discovery Methods..

Interviews…

How to conduct an *Interview*?...

Prepare for the Interview…

Interview question guidelines :

- Use clear and concise language
  - Don’t include your opinion as part of a question
- Avoid long or complex questions
- Avoid threatening questions
- verify before you leave

The purpose of the interview is to investigate, not to evaluate or criticize

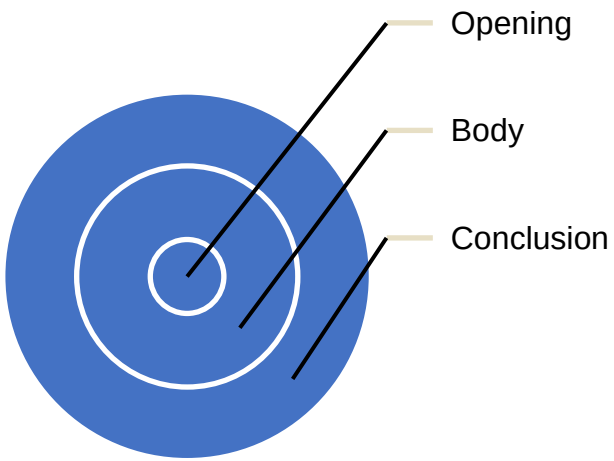
Requirements Discovery Methods..

Interviews…

How to conduct an *Interview*?...

Conduct the Interview

The actual interview consist of three phases:



Interview Opening : Intended to influence or motivate the interviewee to participate
Interview body : Obtain interviewee's response to your list of questions
Interview conclusion : Express your appreciation. Important for maintaining good relationship and trust.

Video Conferencing

Examples of Video Conferencing Software

- Zoom.
- Microsoft Teams.
- Google Hangouts Meet.
- Skype.
- Cisco Webex Meetings.
- BlueJeans Meetings.
- GoToMeeting.
- Jabber.

Requirements Discovery Methods..

Prototyping

- A method used to test or illustrate an idea and build a system in an explorative way.
- Used to discover user requirements
- Allows analyst to quickly create mock forms and tables to simulate the implemented system.

Requirements Discovery Methods..

Prototyping

Advantages

- Allow users and developers to experiment with the software and develop with an understanding
- Helps to determine feasibility and usefulness of the system
- Minimize the time spent for fact-finding and help define more stable requirements. Etc.

Disadvantages

- Developer may need to be trained in the prototyping approach
- Prototype can only simulate system functionality and are incomplete in nature. Etc.

Joint Requirements planning (JRP)

- Provides a working environment in which to accelerate all systems analysis tasks and deliverables
- Promotes enhanced system owner and system user participation in systems analysis
- Requires a facilitator for the workshop with superior mediation and negotiation skills
- Used with model driven approaches
- Incorporated into rapid application development(RAD) methodologies

Requirement Discovery Methods.

Joint Requirement Planning (JRP)

- Highly structured group meetings are conducted to analyze problems and define requirements.
- JRP is a subset of a more comprehensive joint application development or JAD technique

JRP Participants

- Sponsor
  Serve as JRP champion. Single person who is in top management.
- Facilitator
  Single individual who plays the role of the leader or facilitator.

Requirement Discovery Methods…

Joint Requirement Planning (JRP)…

JRP Participants…

- Users and Managers

  Number of participants from the user and management.

- Scribes

  Those who are responsible for keeping records pertaining to everything discussed in the meeting.

- IT Staff

  IT personal who primarily listen and take notes regarding issues and requirements.

For more details refer - Ref_3